



UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT



Software Design

Progress Report No. 7

Designing the System

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I. Objectives

In this section, the goals in this laboratory are:

- Describe the conceptual design and technical design of the Web-Based Salon Management and Booking System.
- Transform the identified system requirements into a structured and workable system design that can be implemented effectively.

II. Methods

What is Design?

Customers request a new system either because no existing system is available or because the current system has limitations that affect efficiency and usability. In both cases, the requirements documentation defines the problems that the system must solve. Design is the creative and systematic process of transforming these problems into a practical solution. The output of this process is a detailed description of how the system will operate and how it will meet user's needs.

Two Part of the Design Process

Conceptual or System design

This design level focuses on describing what the system will do from the user's perspective. It defines system functions, user interactions, data flow, and overall behavior without focusing on specific technologies.

Technical design

This design level focuses on how the system will be built. It specifies the hardware configuration, software architecture, database structures, and technologies required to implement the system.

III. Presentation Content

After the design phase is completed, review meetings are held to make sure the proposed system meets both customer needs and technical requirements before moving forward with development. These reviews are carried out in three stages that follow the design process.

The first stage is the Preliminary Design Review (PDR), where the conceptual design is discussed with customers and users to gather their feedback. This is followed by the Critical Design Review (CDR), which focuses on closely examining the technical design together with system developers. Lastly, a Program Design Review is conducted to give programmers the opportunity to review the design and provide input before the system is fully implemented.

IV. Preliminary Design Review

1. Overview of the Conceptual Design

The Web-Based Salon Management and Booking System is designed to automate and centralize salon operations such as appointment scheduling, customer management, service tracking, and billing. The system allows customers to book appointments online, while salon staff and administrators manage schedules, services, and records through a secure web interface.

The conceptual design focuses on user interaction, system workflows, and data flow to ensure that all client requirements are addressed.

2. System Users and Roles

User Type	Responsibilities
Administrator	Manages services, staff schedules, system settings, and reports
Staff	Views appointments, updates service status, assists customers
Costumer	Registers, logs in, books appointments, views booking history

3. Hardware Configuration

At the conceptual level, the system requires:

- A server to host the web application and database
- Client devices such as desktops, laptops, tablets, or smartphones
- Internet connectivity for system access

4. Software Architecture

The system follows a **web-based client–server architecture**:

- Users interact with the system through a web browser
- The application server processes requests such as booking and data retrieval
- A centralized database stores customer, appointment, service, and transaction data

5. Data Structures

The system manages the following core data entities:

- **Customer** (name, contact details, login credentials)
- **Appointment** (date, time, service, assigned staff)
- **Service** (service name, price, duration)
- **Transaction** (billing details, payment status)

These data structures support scheduling, record management, and reporting.

6. Data Flow (Input–Process–Output)

Input:

- Customer registration details
- Appointment booking requests
- Service updates by staff

Process:

- Validation of appointment availability
- Storage and retrieval of customer records
- Generation of transaction records

Output:

- Appointment confirmations
- Updated schedules for staff
- Reports for salon administrators

7. Validation of Requirements

The preliminary design ensures that:

- All customer requirements identified during analysis are addressed
- The system supports efficient appointment scheduling
- Data is centralized and accessible to authorized users
- The system improves operational efficiency and customer experience

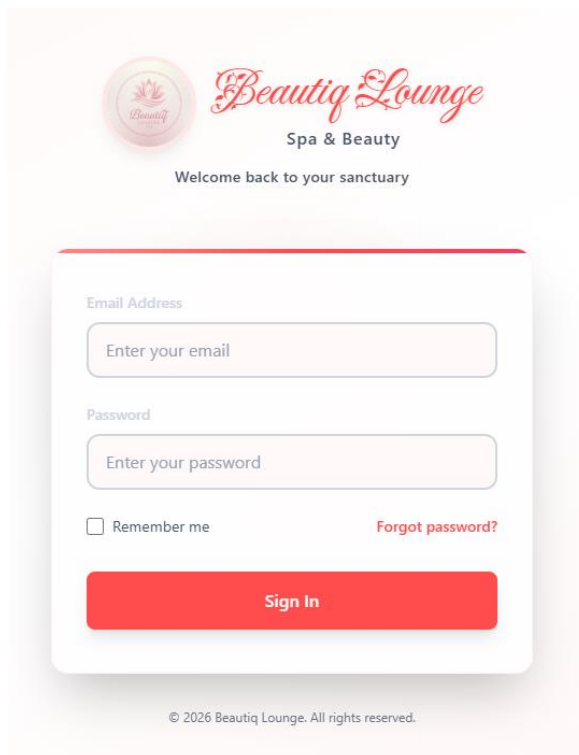


Figure 1. Screenshot of the Proposed Web-Based Salon Management and Booking System

V. Conclusion

This laboratory focused on designing the Web-Based Salon Management and Booking System by turning the identified requirements into a clear and organized system design. Through both conceptual and technical design processes, the system’s main functions, structure, and implementation approach were defined to meet the operational needs of the salon.

The design activities were completed through effective teamwork, with each member sharing ideas and feedback during discussions and review sessions. Working collaboratively helped ensure that the proposed system was practical, easy to understand, and aligned with the client’s requirements. Overall, this laboratory emphasized the importance of proper system design and collaboration in building a strong foundation for successful system development.

References

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